MSHA Finger Pinch/Crush Narrative Comparison

Corrected Focused K-means and LDA Analysis

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Background corpus used for fitting: 2,000
Shared vocabulary terms: 2,763
Selected K-means cluster: 6
Selected cluster size: 247
Matched LDA topic: 3
LDA membership: dominant topic or proportion >= 0.50
LDA topic members: 408
Agreement records: 200
K-means-only records: 47
LDA-only candidates: 208
Cluster capture rate: 81.0%
Focused Jaccard: 0.440

Scope:
All 2,000 narratives were used to fit the models. The report focuses
only on the selected finger cluster, the matched LDA topic, and the
documents for which those methods agree or disagree.
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Executive Summary

This report uses all 2,000 narratives to fit the vocabulary, K-means clusters, and LDA topics, but the analysis is narrowed to one selected finger pinch/crush cluster and its closest LDA topic.

K-means cluster 6 contained
247 narratives and had a finger-event indicator
rate of 100.0%.

LDA topic 3 was selected as the closest topic.
LDA membership was defined independently using the dominant-topic rule
or a topic-proportion threshold of
0.50.

The methods agreed on 200 records. K-means retained
47 records that did not meet the LDA membership rule.
LDA identified 208 records from outside the K-means
cluster as possible related documents.

Agreement records provide the strongest shared interpretation.
LDA-only records are candidates for relevant narratives that K-means may have placed elsewhere, but these records still require qualitative review.

Focused Comparison Method

1. A severity-balanced sample of 2,000 narratives was used to fit both methods.

2. One synchronized vocabulary was created using the same cleaning, stop-word removal, document-frequency limits, and one- and two-word phrases.

3. K-means used TF-IDF and Truncated SVD and assigned each narrative to exactly one of nine clusters.

4. LDA used term counts and represented each document as a mixture of nine topic proportions.

5. The finger pinch/crush K-means cluster was selected from its vocabulary and narrative content.

6. The closest LDA topic was selected using its concentration inside the K-means cluster, separation from outside documents, enrichment, and dominant-topic frequency.

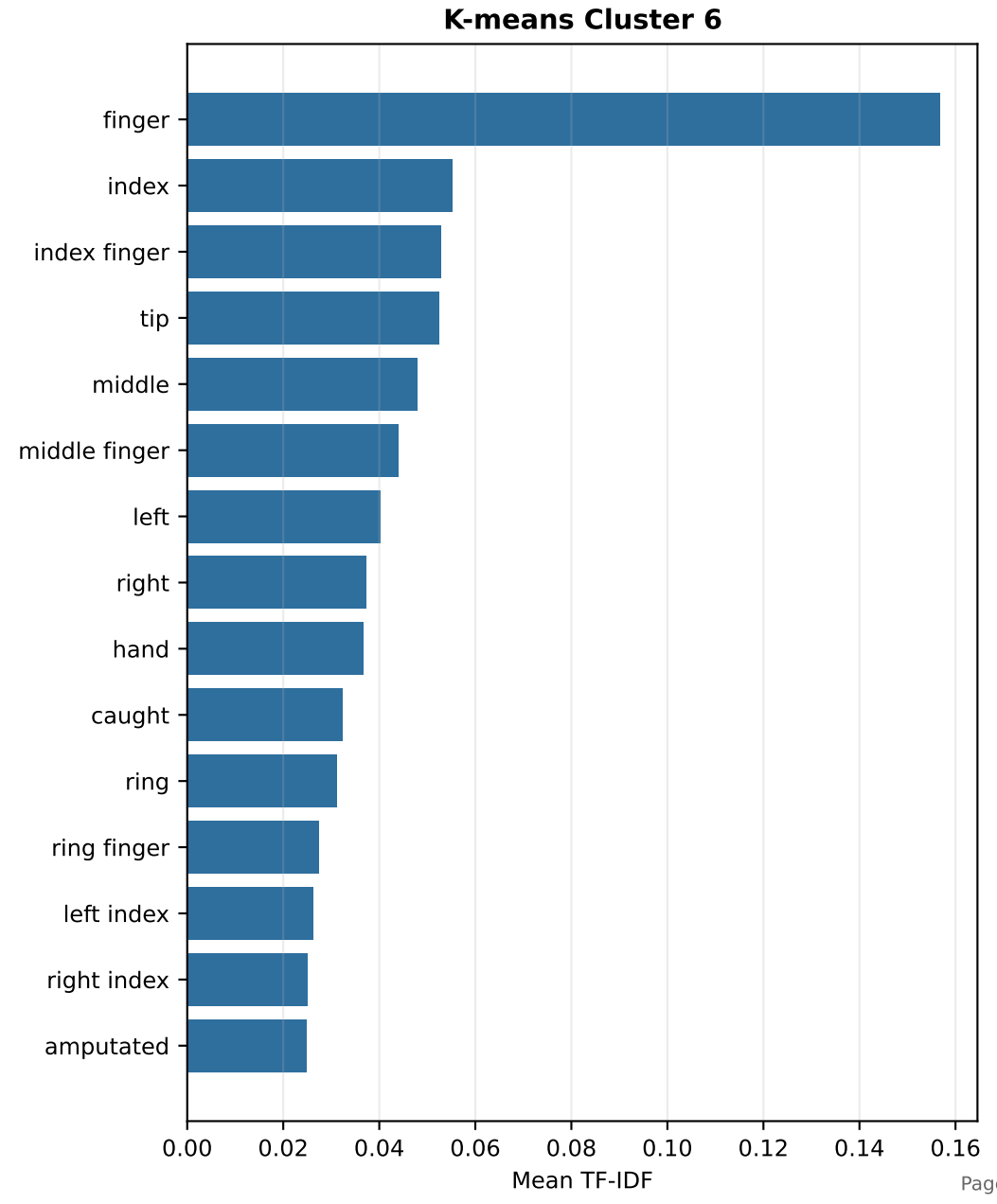
7. LDA membership was based on a topic rule, not a forced equal-size document set.

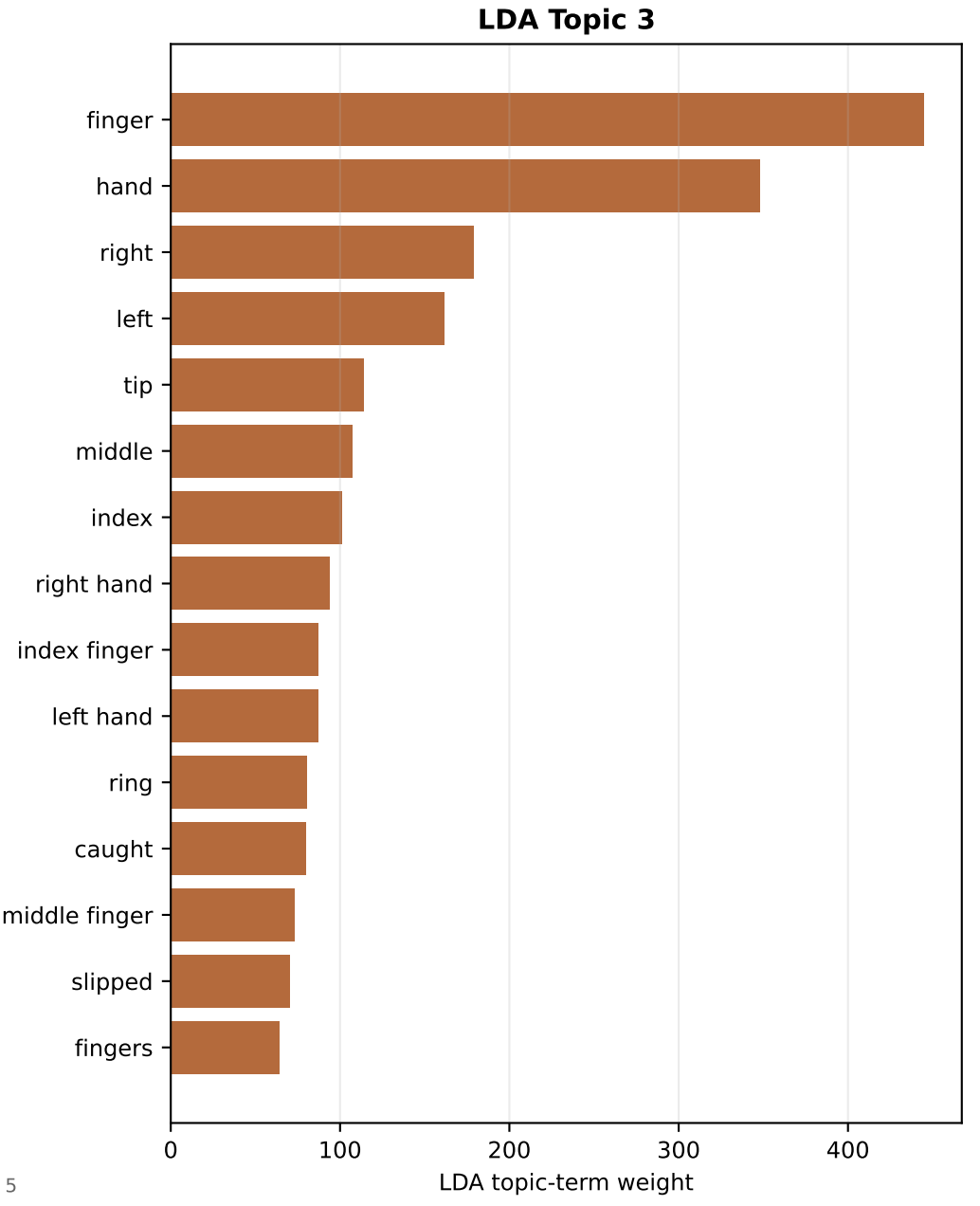
8. The report compares Agreement, K-means-only, and LDA-only records. The remaining corpus is used only as model-fitting background.

Model-Fitting Sample

Severity Group	Available Records	Target	Initially Selected	Final Selected
Fatality	1188	400	400	400
Permanent disability	2455	400	400	400
Days away	178432	400	400	400
Restricted or transferred	42788	400	400	400
No lost time or other	43965	400	400	400

Vocabulary Comparison for the Selected Finger Pattern





Top-Term Comparison

SHARED TERMS
---
caught, finger, hand, index, index finger, left, middle, middle finger, right, ring, tip

K-MEANS-ONLY TERMS
---
amputated, left index, right index, ring finger

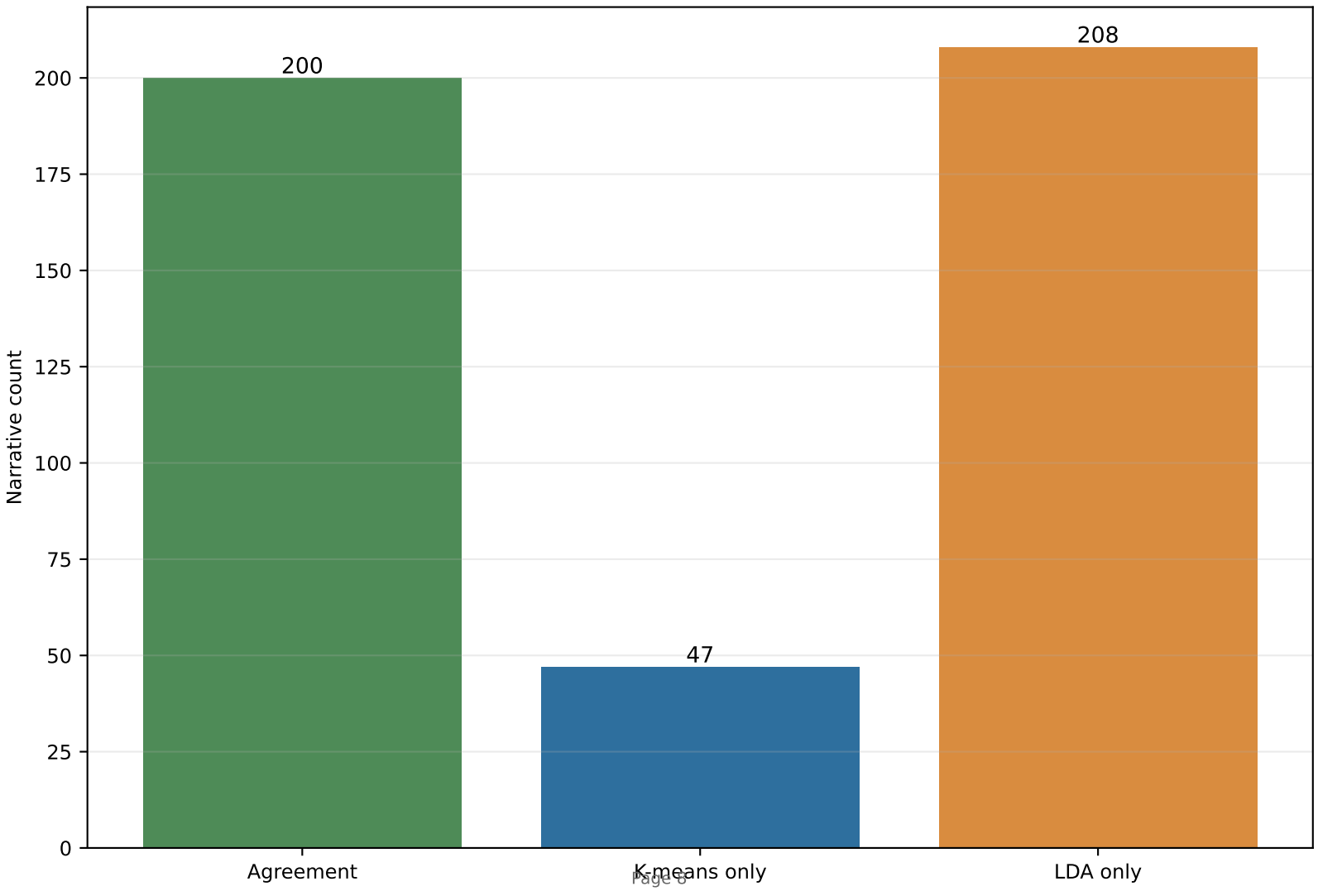
LDA-ONLY TERMS
---
fingers, left hand, right hand, slipped

Top-term Jaccard overlap: 0.579

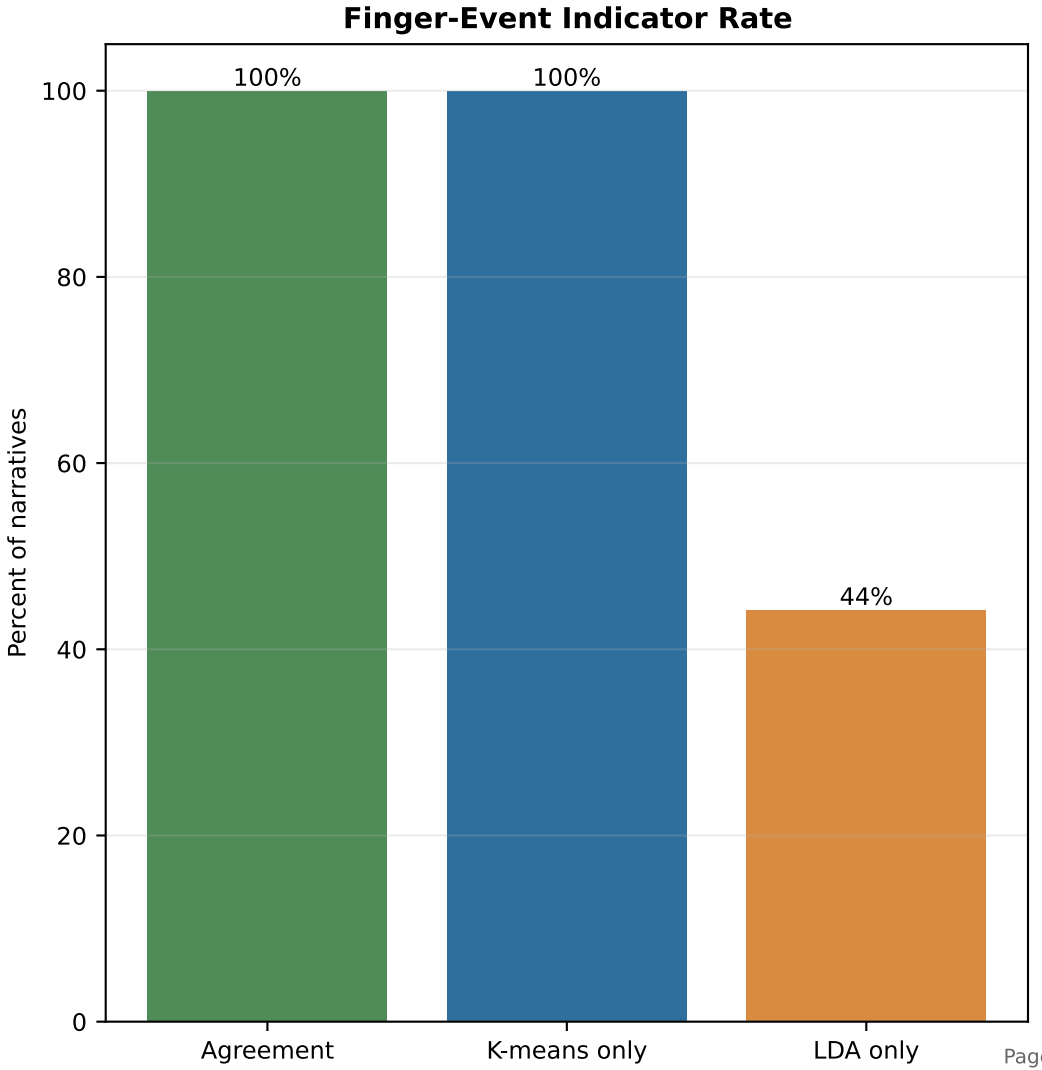
K-means terms identify vocabulary distinguishing the selected cluster.
LDA terms represent vocabulary associated with a probabilistic topic
that may occur across several K-means clusters.

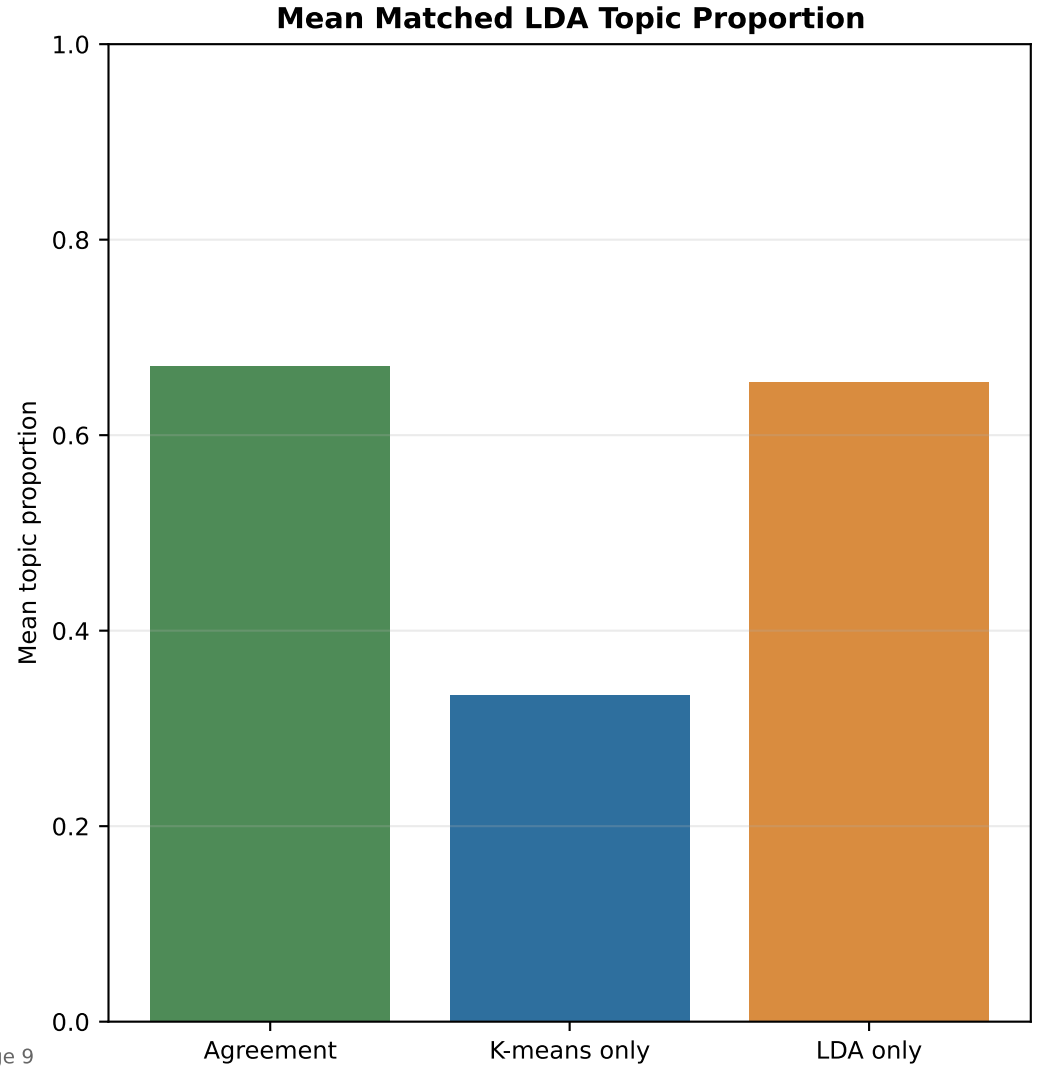
LDA Topic Matching Results

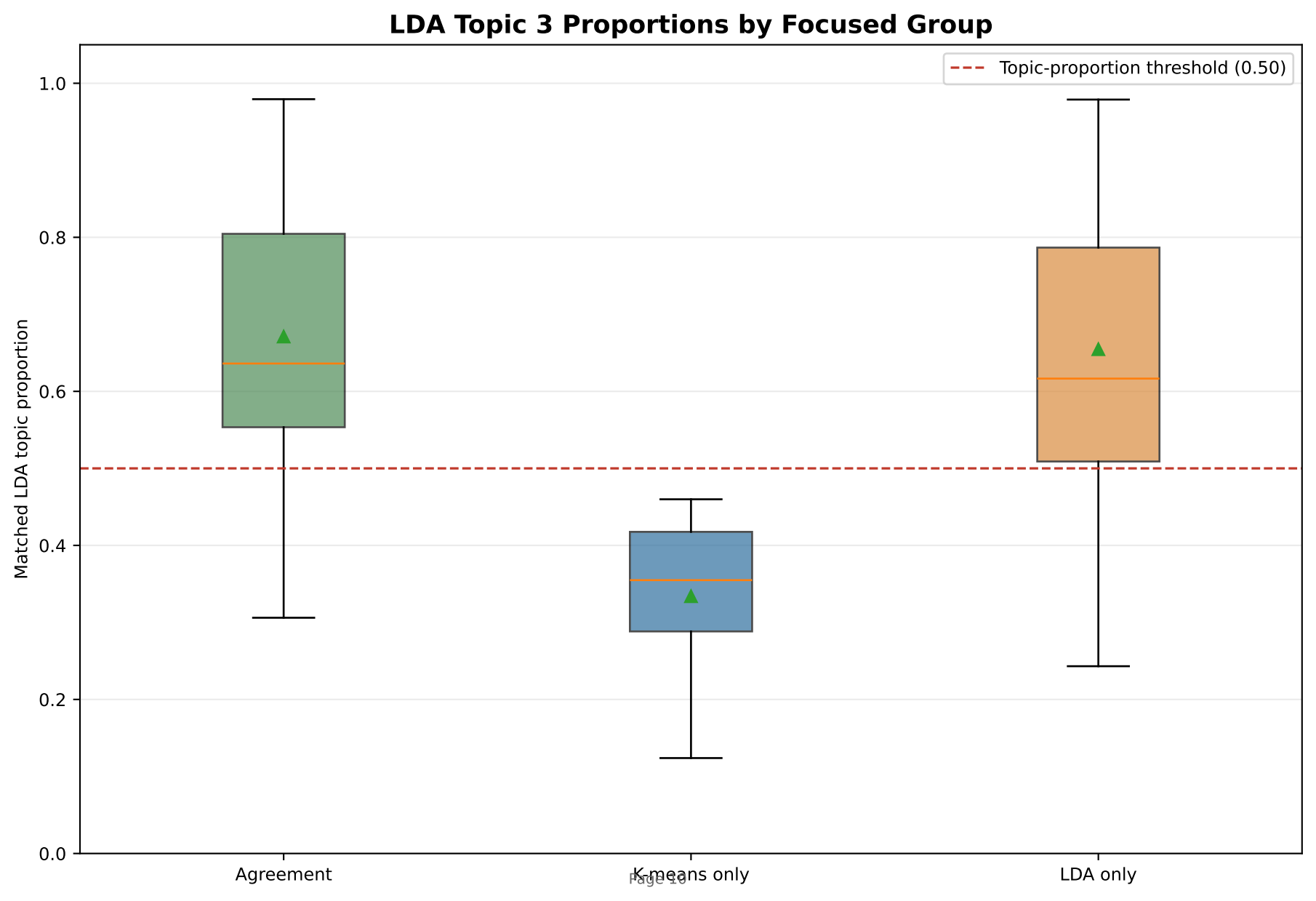
Topic Id	Mean Proportion Inside Clu	Mean Proportion Outside C	Inside-Outside Separation	Enrichment Ratio	Dominant-Topic Rate Inside	Combined Match Score
3	0.6069	0.1165	0.4904	5.2095	0.8097	0.6475
8	0.1373	0.1326	0.0047	1.0357	0.0891	0.1039
2	0.0729	0.1434	-0.0706	0.5081	0.0405	0.0527
1	0.0496	0.1040	-0.0545	0.4764	0.0121	0.0361
7	0.0400	0.0817	-0.0417	0.4894	0.0202	0.0328
6	0.0393	0.1175	-0.0782	0.3343	0.0081	0.0275
0	0.0284	0.1340	-0.1056	0.2122	0.0121	0.0203
4	0.0133	0.0912	-0.0779	0.1456	0.0040	0.0102
5	0.0123	0.0791	-0.0667	0.1561	0.0040	0.0099

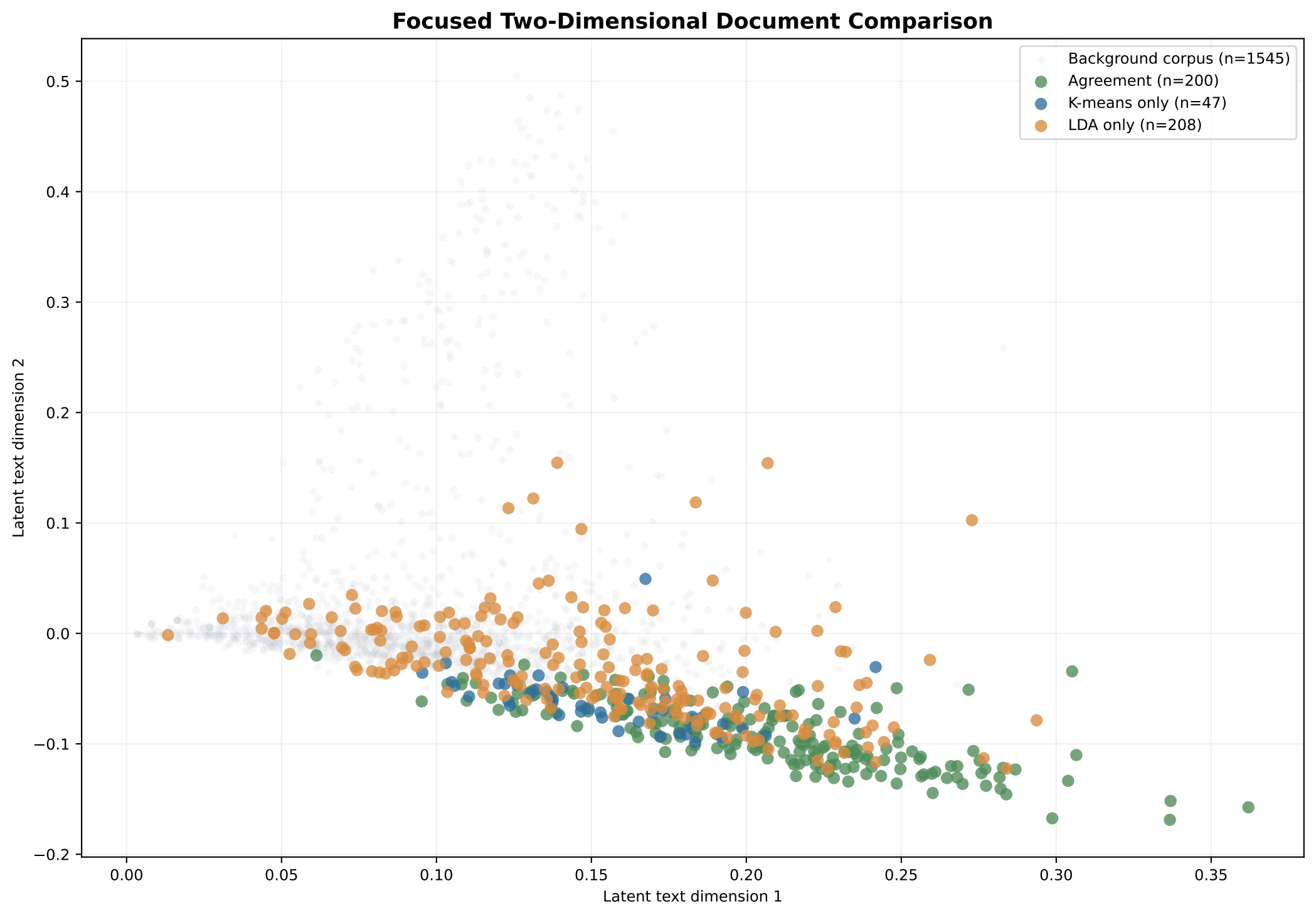

Focused Document Comparison
Narrative count
0
25
50
75
100
125
150
175
200
200
47
208
Agreement
K-means only
LDA only

Focused Group Quality Comparison

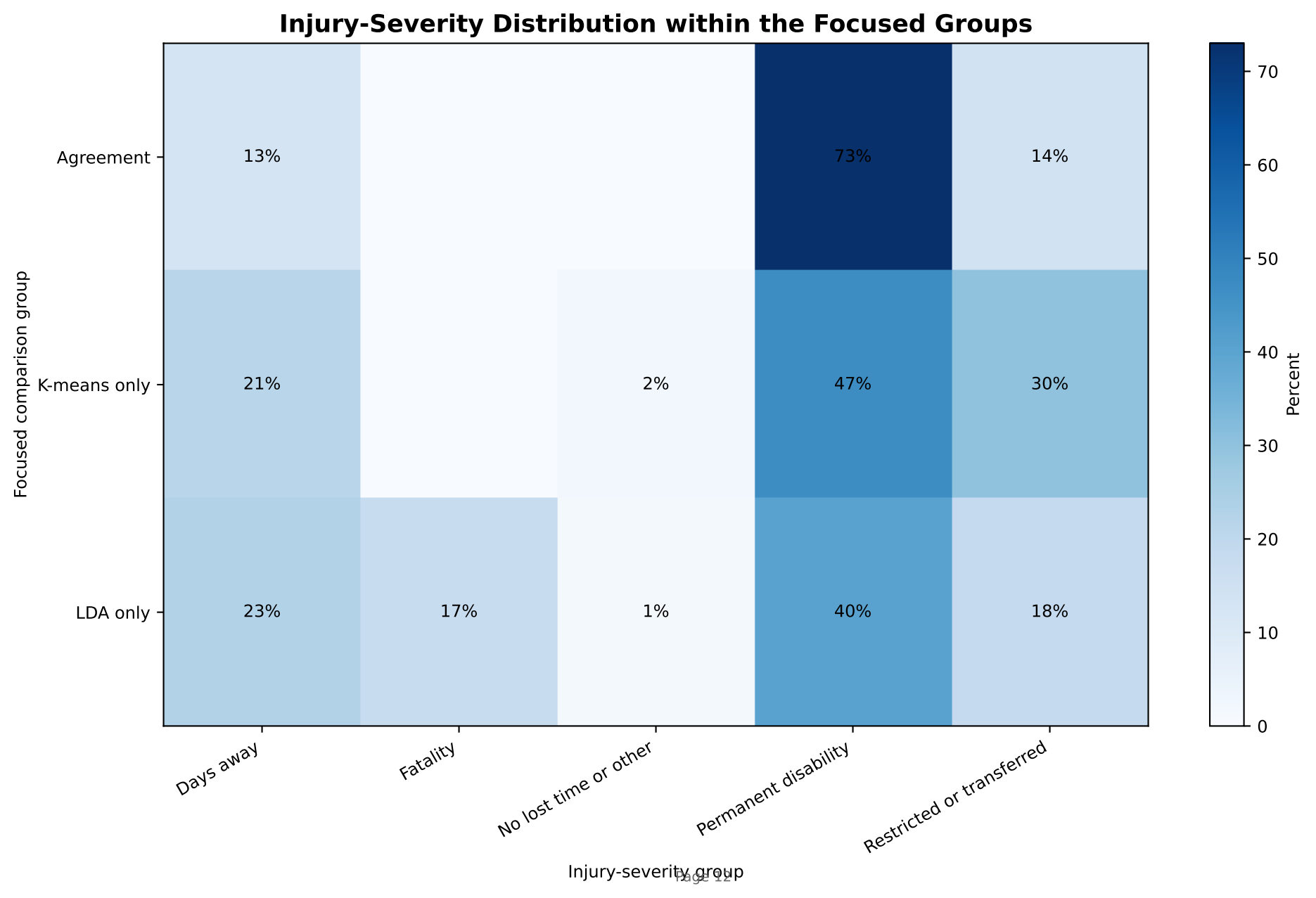





LDA Topic 3 Proportions by Focused Group
Topic-proportion threshold (0.50)
Matched LDA topic proportion
1.0
0.8
0.6
0.4
0.2
0.0
Agreement
K-means only
LDA only



The light-gray documents provide model-fitting context. The colored documents are the focused analytical groups.


Injury-Severity Distribution within the Focused Groups
Focused comparison group
Agreement
K-means only
LDA only
13%
73%
14%
21%
2%
47%
30%
23%
17%
1%
40%
18%
Days away
Fatality
No lost time or other
Permanent disability
Restricted or transferred
Injury-severity group
Percent
70
60
50
40
30
20
10
0

Post-Hoc Interpretation

The existing MSHA fields on the following pages were not used to create the K-means cluster or LDA topic.

The tables are included only to evaluate whether Agreement, K-means-only, and LDA-only records describe similar body parts, injury outcomes, activities, accident types, and classifications.

The most important check is whether the LDA-only records represent finger pinch, crush, caught-between, degloving, or amputation events that K-means placed into another narrative cluster.

Existing MSHA Fields by Focused Group

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
Agreement	CLASSIFICATION	1	HANDLING OF MATERIALS	96	48.0%
Agreement	CLASSIFICATION	2	MACHINERY	55	27.5%
Agreement	CLASSIFICATION	3	HANDTOOLS (NONPOWERED)	22	11.0%
Agreement	CLASSIFICATION	4	POWERED HAULAGE	18	9.0%
Agreement	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	84	42.0%
Agreement	ACCIDENT_TYPE	2	Caught in, under or between NEC	51	25.5%
Agreement	ACCIDENT_TYPE	3	Caught in, under or between running or meshing objects	19	9.5%
Agreement	ACCIDENT_TYPE	4	Struck by... NEC	19	9.5%
Agreement	ACTIVITY	1	Machine maintenance	65	32.5%
Agreement	ACTIVITY	2	Handling supplies or material	49	24.5%
Agreement	ACTIVITY	3	Hand tools (not powered)	23	11.5%
Agreement	ACTIVITY	4	Handling coal, Rock, Ore	8	4.0%
Agreement	INJURY_SOURCE	1	METAL,NEC(PIPE,WIRE,NAIL)	43	21.5%
Agreement	INJURY_SOURCE	2	METAL COVERS & GUARDS	29	14.5%
Agreement	INJURY_SOURCE	3	UNDERGRD MINING MACHINES	10	5.0%
Agreement	INJURY_SOURCE	4	BROKEN ROCK,COAL,ORE,WSTE	10	5.0%
Agreement	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	145	72.5%
Agreement	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	22	11.0%
Agreement	NATURE_INJURY	3	FRACTURE,CHIP	22	11.0%
Agreement	NATURE_INJURY	4	CRUSHING	6	3.0%
Agreement	INJ_BODY_PART	1	FINGER(S)/THUMB	196	98.0%
Agreement	INJ_BODY_PART	2	HAND (NOT WRIST OR FINGERS)	4	2.0%

Existing MSHA Fields by Focused Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
Agreement	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	146	73.0%
Agreement	DEGREE_INJURY	2	DAYS RESTRICTED ACTIVITY ONLY	28	14.0%
Agreement	DEGREE_INJURY	3	NO DYS AWY FRM WRK,NO RSTR ACT	20	10.0%
Agreement	DEGREE_INJURY	4	DAYS AWAY FROM WORK ONLY	5	2.5%
K-means only	CLASSIFICATION	1	HANDLING OF MATERIALS	25	53.2%
K-means only	CLASSIFICATION	2	HANDTOOLS (NONPOWERED)	11	23.4%
K-means only	CLASSIFICATION	3	POWERED HAULAGE	7	14.9%
K-means only	CLASSIFICATION	4	MACHINERY	4	8.5%
K-means only	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	20	42.6%
K-means only	ACCIDENT_TYPE	2	Struck by... NEC	9	19.1%
K-means only	ACCIDENT_TYPE	3	Caught in, under or between NEC	9	19.1%
K-means only	ACCIDENT_TYPE	4	Struck by falling object	5	10.6%
K-means only	ACTIVITY	1	Handling supplies or material	14	29.8%
K-means only	ACTIVITY	2	Machine maintenance	13	27.7%
K-means only	ACTIVITY	3	Hand tools (not powered)	10	21.3%
K-means only	ACTIVITY	4	Handling coal, Rock, Ore	2	4.3%
K-means only	INJURY_SOURCE	1	METAL,NEC(PIPE,WIRE,NAIL)	11	23.4%
K-means only	INJURY_SOURCE	2	METAL COVERS & GUARDS	6	12.8%
K-means only	INJURY_SOURCE	3	BROKEN ROCK,COAL,ORE,WSTE	3	6.4%
K-means only	INJURY_SOURCE	4	AXE,HAMMER,SLEDGE	3	6.4%
K-means only	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	22	46.8%
K-means only	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	17	36.2%

Existing MSHA Fields by Focused Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
K-means only	NATURE_INJURY	3	FRACTURE,CHIP	4	8.5%
K-means only	NATURE_INJURY	4	CRUSHING	1	2.1%
K-means only	INJ_BODY_PART	1	FINGER(S)/THUMB	47	100.0%
K-means only	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	22	46.8%
K-means only	DEGREE_INJURY	2	DAYS RESTRICTED ACTIVITY ONLY	14	29.8%
K-means only	DEGREE_INJURY	3	NO DYS AWY FRM WRK,NO RSTR ACT	10	21.3%
K-means only	DEGREE_INJURY	4	ALL OTHER CASES (INCL 1ST AID)	1	2.1%
LDA only	CLASSIFICATION	1	MACHINERY	57	27.4%
LDA only	CLASSIFICATION	2	HANDLING OF MATERIALS	54	26.0%
LDA only	CLASSIFICATION	3	POWERED HAULAGE	41	19.7%
LDA only	CLASSIFICATION	4	HANDTOOLS (NONPOWERED)	26	12.5%
LDA only	ACCIDENT_TYPE	1	Caught in, under or between a moving and a stationary object	74	35.6%
LDA only	ACCIDENT_TYPE	2	Struck by... NEC	35	16.8%
LDA only	ACCIDENT_TYPE	3	Struck by falling object	28	13.5%
LDA only	ACCIDENT_TYPE	4	Caught in, under or between NEC	19	9.1%
LDA only	ACTIVITY	1	Machine maintenance	51	24.5%
LDA only	ACTIVITY	2	Handling supplies or material	30	14.4%
LDA only	ACTIVITY	3	Hand tools (not powered)	17	8.2%
LDA only	ACTIVITY	4	Roof bolter, NEC	14	6.7%
LDA only	INJURY_SOURCE	1	UNDERGRD MINING MACHINES	23	11.1%
LDA only	INJURY_SOURCE	2	METAL,NEC(PIPE,WIRE,NAIL)	23	11.1%
LDA only	INJURY_SOURCE	3	MINE JEEP,KERSEY,JITNEY	19	9.1%

Existing MSHA Fields by Focused Group - Continued

Comparison Group	Existing Field	Rank	Existing Value	Count	Group Percent
LDA only	INJURY_SOURCE	4	METAL COVERS & GUARDS	16	7.7%
LDA only	NATURE_INJURY	1	AMPUTATION OR ENUCLEATION	80	38.5%
LDA only	NATURE_INJURY	2	CUT,LACER,PUNCT-OPN WOUND	29	13.9%
LDA only	NATURE_INJURY	3	FRACTURE,CHIP	27	13.0%
LDA only	NATURE_INJURY	4	CRUSHING	27	13.0%
LDA only	INJ_BODY_PART	1	FINGER(S)/THUMB	103	49.5%
LDA only	INJ_BODY_PART	2	MULTIPLE PARTS (MORE THAN ONE MAJOR)	25	12.0%
LDA only	INJ_BODY_PART	3	HAND (NOT WRIST OR FINGERS)	20	9.6%
LDA only	INJ_BODY_PART	4	TRUNK, MULTIPLE PARTS	8	3.8%
LDA only	DEGREE_INJURY	1	PERM TOT OR PERM PRTL DISABLTY	84	40.4%
LDA only	DEGREE_INJURY	2	DAYS RESTRICTED ACTIVITY ONLY	38	18.3%
LDA only	DEGREE_INJURY	3	FATALITY	36	17.3%
LDA only	DEGREE_INJURY	4	NO DYS AWY FRM WRK,NO RSTR ACT	29	13.9%

Agreement: Strong Shared Finger-Event Examples - 1 of 2

These narratives were assigned to the K-means finger cluster and also met the matched LDA topic rule.
These records provide the clearest shared definition of the finger pinch/crush pattern.

==========
Example 1 | Record 220230890016 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.194
Matched LDA topic proportion: 0.553 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

A miner and partner were moving a 10"X10" cap. As they were setting the cap in place one of the miners caught the tip of left index finger. The tip of the finger was de-gloved. Surgery was preformed the following day. Amputation to the first knuckle was required.

==========
Example 2 | Record 220002310036 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.160
Matched LDA topic proportion: 0.697 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

ON SOUTH END HALF DECK OF PRESS FLOOR, TAKING DOWN AN AIR BLOW DOWN POT, WHEN IT ROLLED AND JAMMED HIS RIGHT INDEX FINGER AGAINST THE MID BAR OF THE HAND RAIL. DISTAL TIP OF FINGER AMPUTATED.

==========
Example 3 | Record 220112910049 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.182
Matched LDA topic proportion: 0.951 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

He was helping lay track and caught his right index finger and middle finger between the bottom of a track tie and the bottom and cut the tips off each finger.

==========
Example 4 | Record 220171160040 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.164
Matched LDA topic proportion: 0.702 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Employee was putting a bearing on a mixer with hand. The bearing moved quicker then expected and ee smashed finger between the bearing and the mixer. The tip of left index finger was amputated by dr.

==========
Example 5 | Record 220213410008 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.143
Matched LDA topic proportion: 0.886 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Agreement: Strong Shared Finger-Event Examples - 2 of 2

Employee was cleaning inside of a Reed Gunite Pot and got left hand glove caught between feed wheel and rock shear. Severed tip of middle finger and index finger.

==========
Example 6 | Record 220120230023 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.178
Matched LDA topic proportion: 0.597 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Moving 24-10 915 Eimco out of surface diesel shop, pinched left index finger between steering ram and hydrlic hose. Laceration and smashed finger this resulted in amputation of finger tip below the first joint.

==========
Example 7 | Record 220130880012 | Days away
K-means cluster: 6 | K-means silhouette: 0.174
Matched LDA topic proportion: 0.647 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EMPLOYEE WAS LOADING A TOP ROLLER ONTO A SKID STEER WHILE TEARING DOWN OLD BELT LINE AN SMASHED HIS LEFT MIDDLE FINGER BETWEEN THE SKID AND THE ROLLER. LEFT MIDDLE INDEX FINGER BROKEN AT TIP OF FINGER

==========
Example 8 | Record 220001730059 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.172
Matched LDA topic proportion: 0.631 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EMPLOYEE WAS REMOVING SOME OLD LOCKERS FROM EMPLOYEE LOCKER ROOM AND HIT HIS RIGHT INDEX FINGER BECAME CLOSED IN ONE OF THE LOCKERS SEVERING THE TIP OF HIS FINGER.

K-means Only: Cluster Records Not Captured by LDA - 1 of 2

These records are members of the K-means finger cluster but did not have the matched LDA topic as their dominant topic and did not reach the 0.50 threshold.

==
Example 1 | Record 220050750003 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.157
Matched LDA topic proportion: 0.391 | Matched topic dominant: False
Existing classification: MACHINERY
Existing accident type: Caught in, under or between NEC

EE was repairing the pneumatic grease gun, he removed the air line thinking that he had disconnected the energy. He then removed the side covers and began to clean out the grease fron under the piston. While doing this the grease gun fired causing the piston to crush his left index finger behind the first knuckle. The finger tip could not be repaired and was removed.

==
Example 2 | Record 220142870002 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.126
Matched LDA topic proportion: 0.429 | Matched topic dominant: False
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Laborer was manually rotating rotor to the area that was to be fixed. As it was being rotated his hands were placed on top and side of blow bar. As it was being rotated the left ring finger got pinched between blow bar and check plate resulting in taking partial tip of finger. The laborer was wearing leather gloves.

==
Example 3 | Record 220172960021 | Days away
K-means cluster: 6 | K-means silhouette: 0.135
Matched LDA topic proportion: 0.432 | Matched topic dominant: False
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between a moving and a stationary object

Employee pinched right index finger between the gate and the roller while closing the gate causing a laceration that needed two stitches to close wound.

==
Example 4 | Record 220111750032 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.149
Matched LDA topic proportion: 0.268 | Matched topic dominant: False
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Struck by... NEC

Employee loaded shooter with cap - In an attempt to hook lead line employee accidentally shot the shooter injuring his left index finger.

==
Example 5 | Record 220061350004 | Days away
K-means cluster: 6 | K-means silhouette: 0.129
Matched LDA topic proportion: 0.460 | Matched topic dominant: False

K-means Only: Cluster Records Not Captured by LDA - 2 of 2

Existing classification: POWERED HAULAGE
Existing accident type: Struck against stationary object

Employee removed an inspection plug from a rotary feeder while it was still in motion. A short moment later he was distracted and turned his body. When he turned his left index finger entered the inspection plug making contact with the rotating vane assembly. He suffered an avulsion on the tip of his index finder.

==
Example 6 | Record 220070640079 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.093
Matched LDA topic proportion: 0.426 | Matched topic dominant: False
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Struck by falling object

Employee poking lump at cement rock quarry silo # 5. He was prying a lump which dropped causing bar to pinch finger against hand railing. finger tip is swollen and discolored. He went to a Doctor 2/27/07.

==
Example 7 | Record 220063130063 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.109
Matched LDA topic proportion: 0.369 | Matched topic dominant: False
Existing classification: MACHINERY
Existing accident type: Struck by falling object

While pulling core tube out of hole - employee was assisting tube. Overshot let go of the tube causing tube to come down. Employee had his hand on top of chuck severing left index finger.

==
Example 8 | Record 220202870022 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.109
Matched LDA topic proportion: 0.262 | Matched topic dominant: False
Existing classification: HANDLING OF MATERIALS
Existing accident type: Struck against stationary object

Emplooyee claims cut left index finger in Production Shaft on 09/25/2020. Did not report until 10/02/2020 after finger became infected and required treatment. Finger was lanced and antibiotics prescribed. Employee placed on light duty restrictions.

LDA Only: Candidate Finger Events Missing from K-means - 1 of 3

These narratives were assigned to other K-means clusters but met the matched LDA topic rule. These are the primary candidates for related finger events that the first-pass K-means cluster may have missed.

==
Example 1 | Record 220080360005 | Permanent disability
K-means cluster: 7 | K-means silhouette: -0.024
Matched LDA topic proportion: 0.979 | Matched topic dominant: True
Existing classification: POWERED HAULAGE
Existing accident type: Caught in, under or between running or meshing objects

The employee was cleaning out around the Zero Speed hose, while the belt was running. He opened up the guard and stuck his Right hand in, past bearing for the tail pulley, with a pair of Channel locks to clean. When he lost his footing, slipped and fell into the belt, the tail pulley grabbed his channel locks and pulled him into the tail pulley.

==
Example 2 | Record 220041280050 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.021
Matched LDA topic proportion: 0.975 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Caught in, under or between NEC

EE was investigating a noise coming from MC-1 unit, cleaner conveyer tail pulley. He removed the guard from the cleaner conveyer unit and inserted his right hand to check a weld on the pulley to determine if the weld was defective. The tip of the finger was caught and pinched between the bushing and the tail pulley shaft, resulting in loss of fingertip above cuticle.

==
Example 3 | Record 220110540001 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.006
Matched LDA topic proportion: 0.975 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Member was in the deck operating the bolter. He went to reposition himself and stuck his hand on top of the canopy to pull himself up and hit the tram. The bolter moved, smashing his right hand into the top. Member immediately ceased to work and was taken by ambulance to CAMC General where he was told his right hand/middle finger had to be amputated.

==
Example 4 | Record 220253170158 | Restricted or transferred
K-means cluster: 8 | K-means silhouette: 0.046
Matched LDA topic proportion: 0.972 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Injured MTM was sitting in a spare 3rd stage mainframe in the zinc shop installing a horse collar liner. EE was operating the overhead crane with EE's right hand and repositioning a horse collar liner for final placement with EE's left hand when EE's left fingertips became pinched between the bottom of the tub and the liner. This resulted in a fractured left middle fingertip

LDA Only: Candidate Finger Events Missing from K-means - 2 of 3

```
================================================================================
Example 5 | Record 220141350001 | Permanent disability
K-means cluster: 8 | K-means silhouette: -0.000
Matched LDA topic proportion: 0.971 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between running or meshing objects

The employee was cleaning the rotary vane valve unit on a bag house discharge. The employee placed his
hand into the vane valve housing, at which time the unit activated, rotated, and caught the end of a
finger on his left hand, causing a crushing avulsion.

================================================================================
Example 6 | Record 220041950007 | Days away
K-means cluster: 8 | K-means silhouette: -0.027
Matched LDA topic proportion: 0.970 | Matched topic dominant: True
Existing classification: HANDLING OF MATERIALS
Existing accident type: Struck by... NEC

The tailgate on the back of the Mack dump truck would not latch properly.  In order to get tailgate to
latch employee pulled the tailgate up with one hand and lifted the latch with the other.  The tailgate
slipped out of his hand and smashed the middle finger on his left hand.  Employee required 6 stitches.

================================================================================
Example 7 | Record 220062510033 | Permanent disability
K-means cluster: 1 | K-means silhouette: 0.039
Matched LDA topic proportion: 0.970 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Employee was tramming a roof bolting machine towards the face and turning into a right  crosscut.  He
had his right hand outside of the deck area and when the machine turned his hand was caught between
frame and the coal rib, mashing his right hand.

================================================================================
Example 8 | Record 220162530013 | Days away
K-means cluster: 8 | K-means silhouette: 0.007
Matched LDA topic proportion: 0.970 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Struck by... NEC

Working on C side & had left hand behind bracket & took impact tool to run bolt in. Bolt went through
bracket and smashed left small finger. Cause of injury hand placement. Incident didn't involve any
aspect of compliance with regulations/rules/PPE. Miner proficiency may have been a factor.

================================================================================
Example 9 | Record 220080040012 | Permanent disability
K-means cluster: 8 | K-means silhouette: 0.100
Matched LDA topic proportion: 0.968 | Matched topic dominant: True
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Caught in, under or between a moving and a stationary object
```

LDA Only: Candidate Finger Events Missing from K-means - 3 of 3

The mine guys strapped a motor with the wire part of the come-along around the rafter and as they moved the motor the wire kinked and broke dropping the motor; it fell and hit the side of my body and bounced landing on my hand on the railing of the crane smashing my hand and removing two fingers on my right hand.

==========
Example 10 | Record 220160390021 | Restricted or transferred
K-means cluster: 8 | K-means silhouette: 0.056
Matched LDA topic proportion: 0.967 | Matched topic dominant: True
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Caught in, under or between NEC

An Employee was hammering a small scaffold bar from between 2 steel beams. As the bar moved from between the beams Employee's hand slid in and the beam fell 3" onto right hand. Another employee came and helped move the beam, to free hand.

Borderline Members of the K-means Finger Cluster - 1 of 2

These selected-cluster narratives have the lowest K-means silhouette values. They may fit another K-means cluster nearly as well.

==
Example 1 | Record 220013510004 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.021
Matched LDA topic proportion: 0.591 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

EE WAS BOLTING ON MINER. HE WAS ATTEMPTING TO INSERT DRILL STEEL INTO CHUCK AS HE RAISED THE DRILL HEAD. THE STEEL CAUGHT AGAINST THE ROOF & BENT, CATCHING HIS L. LITTLE FINGER BETWEEN THE DRILL STEEL AND THE BOLTER MAST.

==
Example 2 | Record 220051400016 | Days away
K-means cluster: 6 | K-means silhouette: 0.024
Matched LDA topic proportion: 0.944 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

Member, was moving rt. roof bolter when atrs extension came out its slots. When moving it back in place the end came up smashing finger on roof.

==
Example 3 | Record 220001540056 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.028
Matched LDA topic proportion: 0.964 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between a moving and a stationary object

INJURED WAS ON RIGHT SIDE OF BOLTER WHILE OPERATOR WAS TRAMMING FORWARD FOR NEXT ROW OF BOLTS & BOLTER SLID OVER & GOT ON HIS RIGHT HAND MASHINGTHE ENDS OF HIS RING, MIDDLE & POINTER FINGERS. INJURED WAS TAKEN TO HOSPITAL. THE END OF HIS RING & POINTER FINGER (FROM LAST JOINT) WAS REMOVED.

==
Example 4 | Record 220031500007 | Restricted or transferred
K-means cluster: 6 | K-means silhouette: 0.030
Matched LDA topic proportion: 0.256 | Matched topic dominant: False
Existing classification: HANDTOOLS (NONPOWERED)
Existing accident type: Caught in, under or between NEC

THE MAN WAS LOWERING A RATCHET TYPE LIFTING JACK. THE MAN'S FINGER WAS CAUGHT WHEN THE JACK HEAD DROPPED DOWN. LIGHT WORK DUTY STARTED ON 5/19.

==
Example 5 | Record 220052020014 | Permanent disability
K-means cluster: 6 | K-means silhouette: 0.032
Matched LDA topic proportion: 0.754 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Struck by... NEC

Borderline Members of the K-means Finger Cluster - 2 of 2

Tightening a spring bolt and was using an air wrench, slipped and cut some of his pinky finger off.

==========
Example 6 | Record 220240080028 | Days away
K-means cluster: 6 | K-means silhouette: 0.033
Matched LDA topic proportion: 0.979 | Matched topic dominant: True
Existing classification: MACHINERY
Existing accident type: Caught in, under or between running or meshing objects

Employee was bolting with the twin boom bolter and guiding up the monster mat and bolt when EE accidentally hit the clamp lever and got the webbing between EE's thumb and index finger on the left hand caught between the clamp, resulting in a laceration. Compliance with rules/regs, PPE, mining equip/systems not a factor. Miner proficiency may be a factor.

Conclusions and Recommended Interpretation

1. All 2,000 narratives were used to establish the shared vocabulary and fit both models.

2. The substantive comparison is limited to one selected K-means finger cluster and one matched LDA topic.

3. The two methods agreed on 200 narratives.
   These provide the strongest shared examples of the finger-event pattern.

4. K-means retained 47 records that did not meet the LDA membership rule. These may use different event-mechanism, equipment, or injury vocabulary.

5. LDA identified 208 candidate records outside the K-means cluster. These are possible related narratives that K-means placed elsewhere.

6. LDA-only records should not automatically be added to the cluster. Qualitative review is still needed to confirm that the same abstract finger pinch/crush concept is present.

7. K-means creates an exclusive cluster assignment, while LDA allows each narrative to contain a mixture of topics. Some disagreement is therefore expected and informative.

8. The comparison demonstrates why first-pass clusters require review. Similar incidents may use different equipment, activity, mechanism, and injury language.

Technical Appendix

```
Model-fitting narratives: 2,000
Shared vocabulary terms: 2,763
Minimum document frequency: 3
Maximum document frequency: 70%
N-gram range: (1, 2)
SVD components: 75
SVD explained variance: 0.2270
K-means clusters: 9
Overall K-means silhouette: 0.0533
Selected K-means cluster: 6
Selected cluster size: 247
Selected cluster mean silhouette: 0.0918
LDA topics: 9
Matched LDA topic: 3
LDA proportion threshold: 0.50
LDA topic members: 408
Agreement count: 200
K-means-only count: 47
LDA-only count: 208
Focused Jaccard: 0.4396
Top-term Jaccard: 0.5789
Random state: 42

PDF output: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final
Project/MSHA_Finger_Cluster_Focused_KMeans_LDA_Report_Corrected.pdf
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